

Supplemental Information to:

Phylogenomic assessment of microhylid frogs reveals
widespread taxonomic confusion in the Asterophryinae
and establishes the timing of diversification in Australia

Document S1. Figures S1–SX and Tables S1–SX

Contents

Supplementary Information	2
Molecular Sampling	2
Morphological Sampling	2
Morphological Classification	3
Phylogenetics	7

Supplementary Information

Molecular Sampling

Table S1: Table of molecular samples is included in ‘Sampling/Asterophryinae SampleInfo.xlsx’.

Data
Asterophryinae_SampleInfo.xlsx

Morphological Sampling

Below, we provide information about the collection of morphological traits from the literature, as well as summary statistics about the sampling.

Table S2: Morphological measurements of Asterophryinae frogs, following Zweifel (1972; 2000).

Trait	Name	Measurement
SVL	snout-vent length	length from tip of snout to cloacal opening
HLL	hind limb (tibia) length	tibia length, between heel and outer surface of knee
HW	head width	head width at widest point, generally at the level of the tympanum or jaw angle
IN	internarial distance	distance between centers of external nares
EN	eye nare distance	distance between anterior edge of eye opening and center of external naris
EY	eye diameter	distance between anterior and posterior edges of eye opening
THIRD	third finger diameter	width of disc of third finger measured at a right angle to the axis of the digit
FTD	fourth toe diameter	width of disc of fourth toe measured at a right angle to the axis of the digit

Table S3: Number of species (No.Species) and samples (No.Samples) for each genus included in the morphological analyses.

Genus	No.Species	No.Samples
Aphantophryne	3	3
Asterophrys	4	11
Austrochaperina	16	27
Barygenys	3	4
Callulops	16	69
Choerophryne	7	34
Cophixalus	43	216
Copiula	7	26
Gastrophrynoides	2	4
Hylophorbus	10	22
Mantophryne	2	4
Oreophryne	9	48
Paedophryne	4	18
Siamophryne	1	11
Sphenophryne	5	6
Vietnamophryne	3	3
Xenorhina	8	26
NA	0	0

Morphological Classification

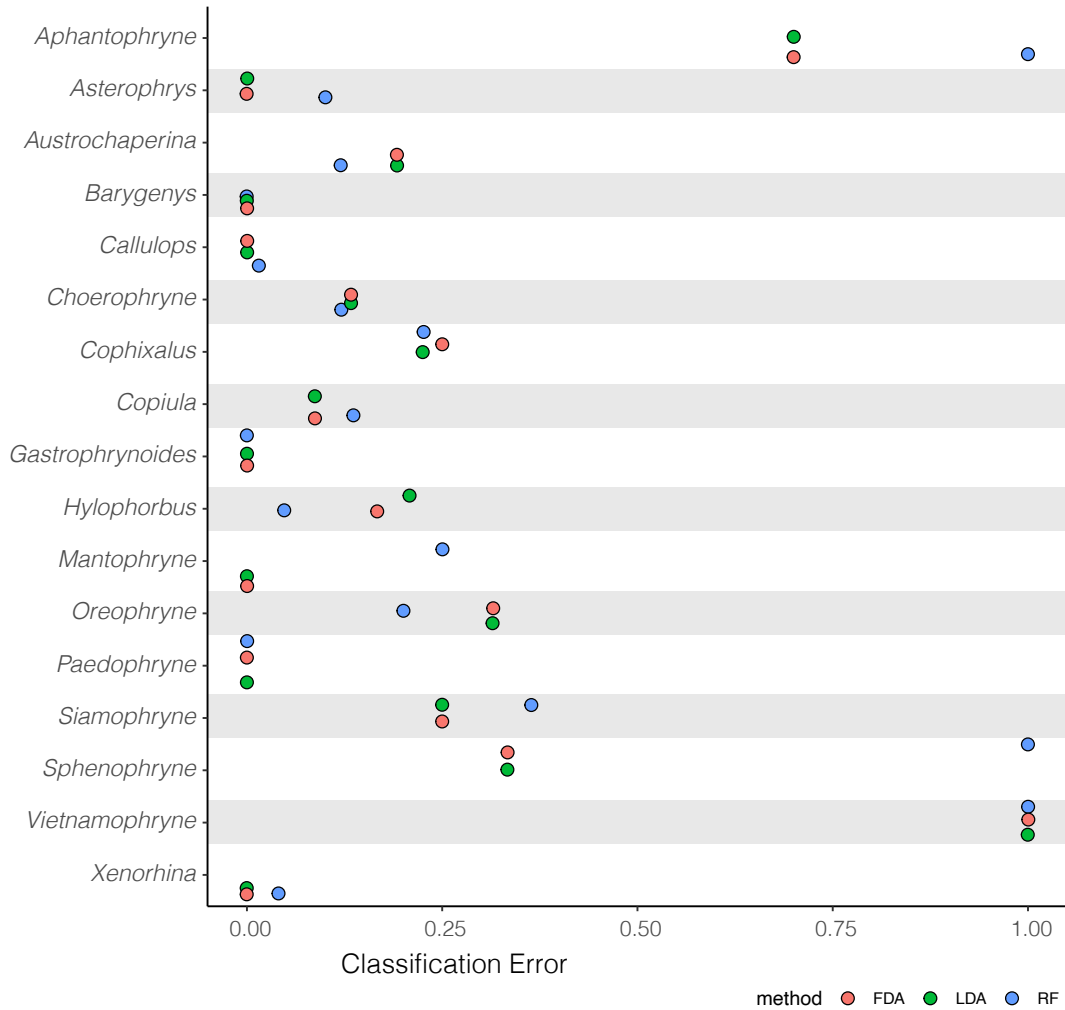


Figure S1. Classification error by Genus and classification method (color). Difference classification methods (FDA–Flexible Discriminant Analysis; LDA–Linear Discriminant Analysis; RF–Random Forest) show comparable levels of classification error and are typically low, with the exception of some genera.

Table S4: Summary of pairwise model classifications across all three classification methods. Table is summarised as the actual genus of the sample (True Genus), the genus it was classified to (Assigned To), and the frequency with which individuals of that genus were classified to that specific other genus (Frequency), depending on method. Misclassifications are highlighted in red.

True Genus	Assigned To	Frequency_RF	Frequency_LDA	Frequency_MDA
Aphantophryne	Aphantophryne	0	3	3
Aphantophryne	Cophixalus	2	5	5
Aphantophryne	Paedophryne	1	0	0
Aphantophryne	Sphenophryne	0	2	2
Asterophrys	Asterophrys	9	7	7
Asterophrys	Callulops	1	0	0
Austrochaperina	Austrochaperina	22	21	21
Austrochaperina	Callulops	0	1	1
Austrochaperina	Choerophryne	0	1	1
Austrochaperina	Cophixalus	2	0	0
Austrochaperina	Copiula	1	1	1
Austrochaperina	Hylophorbus	0	1	1
Austrochaperina	Vietnamophryne	0	1	1
Barygenys	Barygenys	3	3	3
Callulops	Austrochaperina	1	0	0
Callulops	Callulops	66	66	66
Choerophryne	Choerophryne	29	26	26
Choerophryne	Cophixalus	1	1	1
Choerophryne	Hylophorbus	1	0	0
Choerophryne	Oreophryne	2	3	3
Cophixalus	Aphantophryne	2	0	0
Cophixalus	Asterophrys	1	0	0
Cophixalus	Austrochaperina	1	2	2
Cophixalus	Choerophryne	1	0	0
Cophixalus	Cophixalus	41	31	30
Cophixalus	Oreophryne	5	5	5
Cophixalus	Siamophryne	2	2	2
Cophixalus	Vietnamophryne	0	1	1
Copiula	Austrochaperina	3	1	1
Copiula	Copiula	19	21	21
Copiula	Hylophorbus	0	1	0
Copiula	Sphenophryne	0	1	1
Gastrophrynoides	Gastrophrynoides	4	4	4
Hylophorbus	Austrochaperina	1	0	0
Hylophorbus	Cophixalus	0	2	2
Hylophorbus	Hylophorbus	20	19	20
Hylophorbus	Sphenophryne	0	1	1
Hylophorbus	Vietnamophryne	0	1	1
Mantophryne	Callulops	1	0	0
Mantophryne	Mantophryne	3	4	4
Oreophryne	Asterophrys	0	3	3
Oreophryne	Choerophryne	2	5	5
Oreophryne	Cophixalus	7	9	9
Oreophryne	Oreophryne	36	37	37
Paedophryne	Paedophryne	18	18	18
Siamophryne	Cophixalus	3	2	3
Siamophryne	Oreophryne	1	0	0
Siamophryne	Siamophryne	7	9	9
Sphenophryne	Aphantophryne	1	0	0
Sphenophryne	Austrochaperina	1	0	0
Sphenophryne	Callulops	1	0	0
Sphenophryne	Choerophryne	0	1	1
Sphenophryne	Cophixalus	1	0	0
Sphenophryne	Copiula	1	0	0
Sphenophryne	Hylophorbus	1	0	0
Sphenophryne	Sphenophryne	0	2	2
Vietnamophryne	Austrochaperina	1	1	1
Vietnamophryne	Cophixalus	2	3	3
Xenorhina	Callulops	1	0	0
Xenorhina	Xenorhina	24	25	25

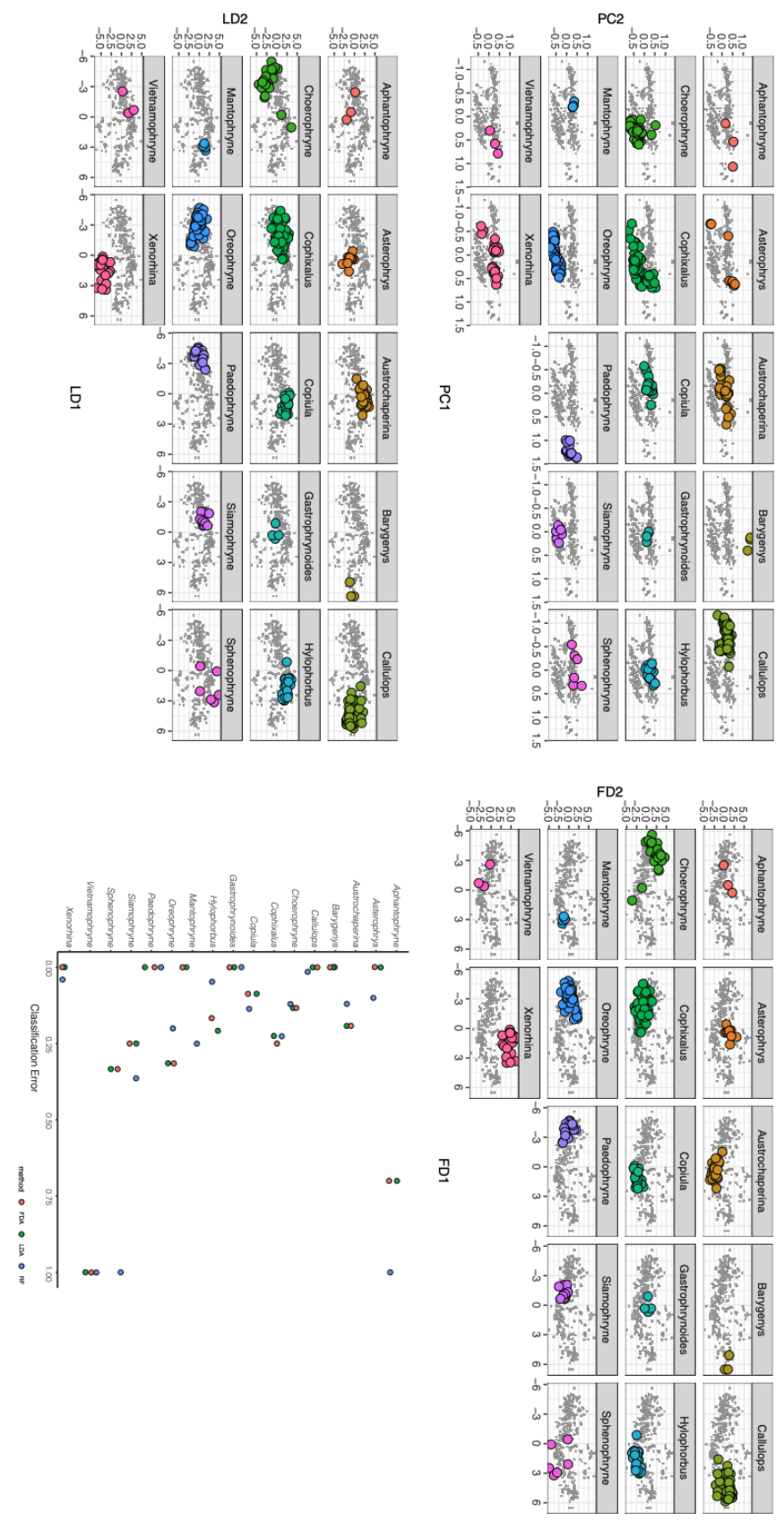


Figure S2. Comparison of visualization of dimensionality reduction techniques.

Error and Bias

Measurement error and bias can be introduced to studies through a number of ways, including inter/intraobserver bias and lack of clarity around how measurements are taken. Haytek et al. (2001) highlights some of these sensitivities, particularly in inter/intraobserver bias. As a result, measurement error can go on to affect downstream interpretations. Our dataset comprises measurements taken across multiple decades by up to 13 different observers, leaving little ability to measure consistency, accuracy, and precision. Getting a single observer to re-measure dozens of individuals 20 or more times for hundreds of species is beyond the means of this study. In Haytek et al. (2001), the focus of the paper is sensitivity in intraspecific sampling. While we include intraspecific sampling in our dataset (none of which were originally published with supporting measurement error), our focus is above the species level. In this case the expectation is that interspecific variation (including error) is greater than intraspecific variation (including error), and that any error is random. To test this idea, we randomly downsampled our morphological dataset to 11 species with more than five individuals measured per species (total 94 individuals). We used this dataset to run a separate Random Forest analysis where the classifier was the species. Of these 11 species, only one (*Cophixalus balbus*) had a classification error > 0 . Looking closer, this is driven by a single individual which is not confidently assigned to the correct species (< 0.70). Given that of 94 individuals across 11 species, only a single individual is misidentified, we can feel relatively confident that our assumption that interspecific (and inter-generic) variation is greater than intraspecific variation is true. We believe this provides evidence of the reliability of the data for our purpose.

Phylogenetics

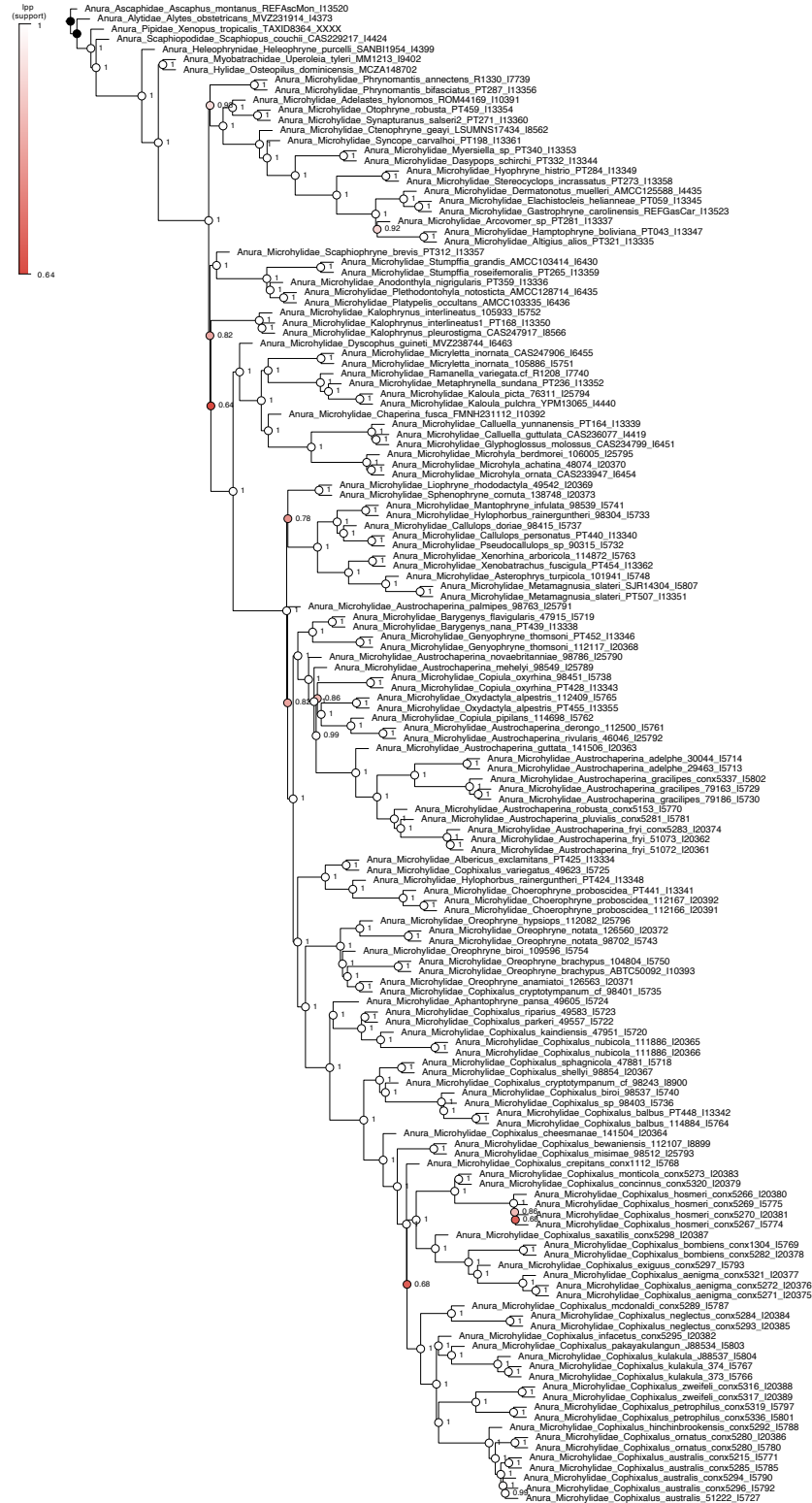


Figure S3. Fully sampled sequence capture tree, estimated by weighted hybrid ASTRAL using IQTREE genetree inputs. Branch support values are indicated at nodes and colored according to value (white = 1; red ≤ 0.9).

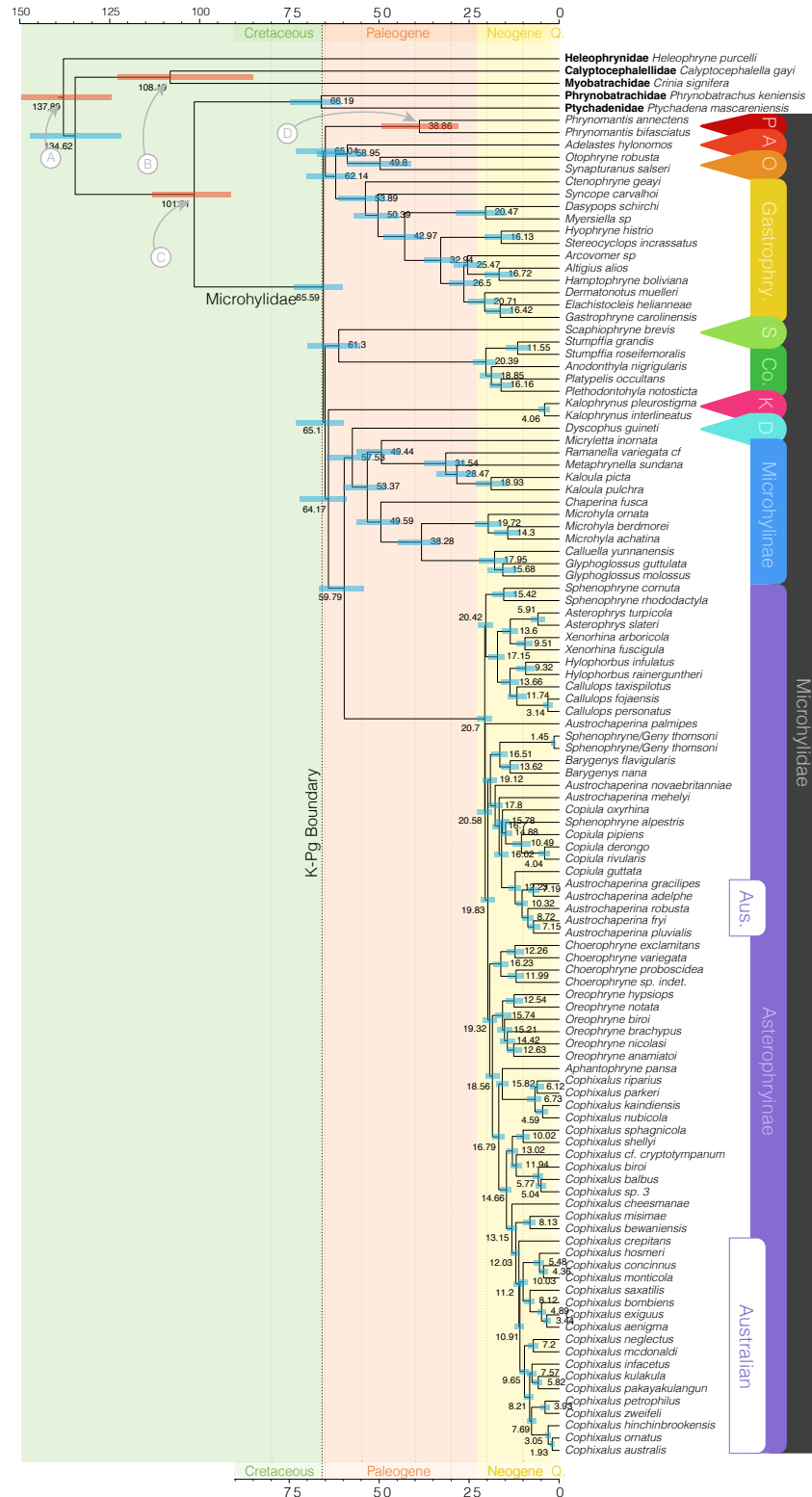


Figure S4. Time calibrated species tree of microhylid frogs highlights with confidence intervals indicated at nodes. Orange colored bars annotated with a circle indicate nodes calibrated by fossil evidence. These correspond to (A) *Beelzebufo ampinga* as a 66 million year minimum on the crown divergence of Neobatrachia; (B) *Calyptocephalella pichileufensis* as a 47.5 million year minimum on the split between Calyptocephalellidae and Myobatrachidae; (C) *Thamatosaurus gezei* as a 33.9 million year minimum on the crown of Ranoidea; and (D) Ptychadenidae fossil as a 25 million year minimum on the split of Ptychadenidae and Phrynobatrachidae.



Figure S5. IQTREE gene tree of concatenated mitochondrial loci (CYTB, ND4). Newly assembled and placed samples are indicated by orange text. Incorporating new samples into the alignments of Hill et al. (2023) allows for a shared understanding of Asterophryinae taxonomy between mitochondrial and nuclear datasets.